

Department of Mathematics, IIT Madras
Scientific Computing Advanced Programming Lab (MA-5105)

QUIZ-I

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- This question paper has four questions, totalling to 10 marks and all are to be answered.

Q.1 A software package is being developed to represent different types of employees in a numerical research laboratory.

A `Researcher` has a name, employee ID, and basic salary, and the program should be able to calculate their total salary. A `TeachingAssistant` has the same basic information and salary, but also has the number of teaching hours and an hourly teaching allowance. A `Professor` also has the same basic information and salary, but additionally receives a research allowance.

Design an object-oriented solution for this problem.

- (a) Identify the three classes and list their important attributes and methods. [1 mark]
- (b) Explain how you would organize these classes to avoid unnecessary repetition of common attributes and methods. [1 marks]

Q.2 A program is required to simulate the motion of projectiles. Each projectile has its own initial position, speed, and launch angle. The program should calculate its position and velocity at any time and determine when it reaches the ground. A separate component should manage the simulation over a specified time interval.

- (a) Identify 2–3 suitable classes. [1 mark]
- (b) For each class, specify its main attributes and methods. [0.5 mark]
- (c) Show the relationship between the classes using a simple class diagram. [0.5 mark]

Q.3 Design a small object-oriented system for solving a system of linear equations using Gaussian elimination. The system should represent matrices, perform row operations such as row exchange and elimination, and solve a linear system given a coefficient matrix and right-hand-side vector.

- ✓(a) Identify 2–3 suitable classes. [1 mark]
- ✓(b) For each class, specify its main attributes and methods. [1 mark]
- ✓(c) Briefly describe how the classes interact during the solution process. [1 mark]

Do not write Python code. Focus on identifying the classes and designing their responsibilities.

✓Q.4 Consider the following Python class:

```
class Vector:
    def __init__(self, x, y):
        self.x = x
        self.y = y

    def __mul__(self, other):
        return Vector(self.x * other, self.y * other)
```

Let

`v = Vector(2, 3).`

For each of the following expressions, state whether it will work. If it works, give the resulting vector; if it does not, explain why.

- ✓(a) `v * 4` [0.5 mark]
- ✓(b) `4 * v` [0.5 mark]
- ✓(c) What change to the class would allow both expressions to work? [2 marks]